

A redescription of *Potamogeton distinctus* including remarks on the taxonomy of the *Potamogeton nodosus* group

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Abstract: *Potamogeton distinctus* has an E. Asian distribution and is closely related to *P. nodosus*. *P. tepperi* seems to be a synonym of *P. tricarinatus*, even though the name has frequently been applied to other species like *P. distinctus*. *P. indicus* is a synonym of *P. nodosus* notwithstanding the fact that not all Indian broad-leaved pondweeds belong to that taxon.

Potamogeton distinctus BENNETT is a species the occurrence of which is well confirmed in Japan and that is therefore recognized by all modern Japanese authors (MIKI 1937, OHWI 1984, KITAMURA & al. 1987). The original description, however, came from the Asian mainland (BENNETT 1904). From that area the species is also mentioned by YUZEPCCHUK (1968), CUONG & VIDAL (1983), YEN (1983) and Botanical Institute (1983). It was included by YANG (1978) in the treatment of the flora of Taiwan. Despite the repeated treatment in floras the exact distribution and the delimitation to species like *P. nodosus* POIRET, *P. polygonifolius* POURRET and *P. wrightii* MORONG is still uncertain. This results from the fact that BENNETT's description is not very precise and, on principle, may refer to any heterophyllous broad-leaved *Potamogeton* species.

Further difficulties arise from the recognition of the taxa *P. delavayi* BENNETT, *P. tepperi* BENNETT and *P. indicus* ROXBURGH. Most of the synonyms and misidentifications of *P. distinctus* are based on errors of early students of *Potamogeton* in times of restricted availability of specimens. However, behind *P. tepperi*, *P. indicus* and *P. delavayi* real taxonomic problems are hidden. That is why these taxa need special consideration.

The aims of the present paper are to present a taxonomical evaluation of the species *P. distinctus* based on field experience and herbarium studies, and to elucidate the structure of the *Potamogeton nodosus* group (WIEGLEB 1988) to which *P. distinctus* has close affinities.

Material and methods

Potamogeton distinctus was collected in 25 sites in Japan. For a comprehensive overview, material of the taxa mentioned above from the following herbaria was studied: A, AAU, ABD, B, BGK, BM, BREM, BRI, C, CANB, CGE, E, F, GAT, GOET, L, LD, M, MICH, MO, P, PH, S, UC, UPNG, UPS, TAI, TI, WU, Z, ZT.

For the study of stem anatomy short pieces of the stems were cut off the specimen, soaked in an ethanol/glycerol mixture for 3 days and proceeded the following way: The pieces were imbedded in hard plastic (Technovit 7 100 and 3 040, Fa. Kulzer). Cross sections of 5 µm were cut with a microtome (Fa. Reichert-Jung) and stained with toluidine blue according to Sakai (GERLACH 1977). Photographs were obtained using a Polyvar research microscope at magnification $\times 100-250$.

Results

1. *Potamogeton distinctus* BENNETT

A. BENNETT, J. Bot. **42**: 72. 1904; GRAEBNER apud ENGLER, Pflanzenreich **4**, **11**: 84. 1907; MIKI, Water Phan. Japan: 17. 1937; A. CAMUS, Fl. Gen. Indochine **6**: 1221. 1942; MASAMUNE, List Vasc. Pl. Taiwan: 131. 1954; MASAMUNE, Enum. Trach. **8**: 203. 1956; YUZEPCCHUK, Fl. U.R.R.S. **1**: 203. 1968; HATUSIMA, Fl. Ryukyus: 644. 1975; YANG, Fl. Taiwan **5**: 28. 1978; Bot. Inst. Chin. Acad. Sci.: 48. 1983; YEN, Chin. Water Pl.: 197. 1983; OHWI, Fl. Japan: 121. 1984; KITAMURA, MURATA & KOYAMA, Col. Ill. Herb. Pl. Jap. **3**: 412. 1987.

Type: Mandschouria, prope Zizikar (Qiqihaer), LITWINOV 2424 (Lectotype, assigned by CUONG & VIDAL 1983, in BM).

≡ *P. distinctus* var. *distinctus* CUONG & VIDAL, Fl. Cambodge, Laos, Vietnam **20**: 58. 1983. p.p. (*P. distinctus* var. *distinctus* is based on the same type as *P. distinctus*, but the Vietnamese specimens listed by CUONG & VIDAL are dubious).

= *P. digynus* WALLICH, Numer. List: 181, no. 5177. 1832; ? YUZEPCCHUK, Fl. U.R.R.S. **1**: 199. 1968.

= ? *P. delavayi* BENNETT, J. Bot. **30**: 228. 1892; ? GRAEBNER apud ENGLER, Pflanzenreich **4**, **2**: 57. 1907 p.p. Type: Eaux courants des fosses dans la plaine de Tali (Da-li), Prov. of Yunnan, China. DELAVAY 4750. 1888 (Holotype in P). (Applications of *P. delavayi* partly refer to *P. nodosus* POIRET, see below.)

= *P. franchetii* BENNETT & BAAGOE, J. Bot. **45**: 234. 1907 p.p.; GRAEBNER apud ENGLER, Pflanzenreich **4**, **11**: 67. 1907; HAGSTROEM, Kungl. Svensk Vetensk. Acad. Handl. **55**, **5**: 156, 1916; HULTEN, Fl. Kamtschatka Adj. Isl.: 76. 1927; MIYABE & KUDO, Fl. Hokkaido Saghalien **2**: 88. 1931; MAKINO & NEMOTO, Fl. Jap. edn. 2: 1282; YUZEPCCHUK, Fl. U.R.R.S. **1**: 198. 1968; Type: Japan, Yokohama. MAXIMOVICZ. 1862 (Holotype in S). (*P. franchetii* partly refers to specimens that have later been named *P. "insularis"* by NAKAI (in herb.). These specimens belong to *P. fryeri* BENNETT.)

= *P. longipetiolatus* CAMUS apud LECOMTE, Not. Syst. **1**: 87. 1910. Type: Coree: pres Chinnampo, juillet 1906, n. 223, rivière Hpyeng-yang, n. 222. FAURIE. Holotype in P.

= *P. tepperi* BENNETT var. α *attenuatus* CAMUS apud LECOMTE, Not. Syst. **1**: 85. 1910. Type: Archipel coreen, OLDHAM 824. Holotype in P.

= *P. tepperi* BENNETT var. β *subcordatus* CAMUS apud LECOMTE, Not. Syst. **1**: 85. 1910. p.p. Type: Ile de Sado, 27 juin 1888, FAURIE 2450. Holotype in P. (The specimen DUCLOUX 2571 is regarded as *P. nodosus*.)

KUDO, Fl. Hokkaido Saghalien 2: 88. 1931; MAKINO & NEMOTO, Fl. Jap. edn. 2: ROSARIO, Philipp. Aquat. Flow. Pl. Ferns: 41. 1967; Type: none. (The specimens listed by BENNETT include *P. distinctus*, *P. cf. nodosus*, and a taxon that was later described as *P. pleiophyllus* HAGSTROEM, which can be best regarded as a *P. natans* hybrid).

= *P. alatus* KOIDZUMI, Bot. Mag. Tokyo 43: 397. 1929. Type: Japonia, prov. Yamashiro, Oguraike (Holotype in TI?).

Also the name *P. elegans* (WALLICH, Numer. List: 181, no. 5178. 1832) may refer to this species. Likewise, the following misidentifications refer to *P. distinctus*:

- *P. natans* sensu MASAMUNE, Enum. Trach. 8: 204. 1956.
- *P. natans* L. var. *polygonifolius* sensu MAKINO, Bot. Mag. Tokyo 1: 3. 1889.
- *P. polygonifolius* sensu FRANCHET & SAVATIER, Enum. Pl. Jap. 2: 14. 1879; BENNETT, Bull. Herb. Boiss. 4: 546. 1896; GRAEBNER in ENGLER, Pflanzenreich 4, 2: 66, p.p.
- *P. tepperi* (non BENNETT 1887) sensu BENNETT, Bull. Herb. Boiss. 4: 548. 1896 (nomen solum); GRAEBNER apud ENGLER, Pflanzenreich 4, 11: 62. 1907 p.p.; MATSUMOTO & HAYATA, J. Coll. Sci. Univ. Tokyo 22: 465. 1906; KITAGAWA, Lin. Fl. Mansh.: 53. 1939; ?KAK, J. Bombay Nat. Hist. Soc. 80: 313. 1983 p.p.
- *P. indicus* sensu BENNETT, Bull. Herb. Boiss. 4: 548. 1896 (nomen solum); GRAEBNER, in ENGLER, Pflanzenreich 4, 2: 64. 1907, p.p.

The official Japanese name is: hiru-mushiro.

Perennial or annual freshwater aquatic herb. The individual plant can be regarded as a shoot complex born from a turion (multiple winterbud) or seed. Lower horizontal shoot (“rhizome”) inserted at the 5th to 7th node of the main shoot, alternate, long creeping (up to 2 m per growth season), whitish, producing up to 15 generations of vertical shoots per season, proliferate branching frequent, producing LHS of up to 4th order. Turion multiple (figures in KADONO 1984: Fig. 7, 1988: photograph), with 2 to 5 dormant buds, slightly thickened, but not really tuber-like as in *P. wrightii*, similar to the turion of *P. nodosus*.

Stem of vertical shoot normally up to 130 cm long (rarely exceeding 2 m), with internodes up to 205 mm, diameter 0.5–1.8 mm, mostly unbranched, very rarely with non-flowering axillary leafy shoots, frequently with short upper horizontal shoots bearing one rooting vertical shoot.

Submersed leaves present or absent, alternate, of variable shape, mostly lanceolate, with acute apex, margin minutely dentate, length 45–140 mm, width (5) 10–23 mm, petiole length 30–190 (230) mm. Floating leaves always present, alternate or subopposite, shape very variable from long lanceolate with acute tip to obovoid with rounded tip, length (30) 60–95 (125) mm, width 10–35 mm, petiole length (48) 80–255 (390) mm, the longest petioles are found in alternate leaves below the flowering region. Phyllodial and stipulate leaves sometimes present at the bases of vertical shoots (KADONO 1984: Fig. 7). Leaves of landforms 40–70 (100) mm long and 15–35 (50) mm wide, oblong to roundish. Stipules 42–85 mm long, acute, persistent.

Peduncles 45–105 mm long, 1.1–2.5 mm wide, mostly slightly thicker than stem, often curved. Spikes 25–80 mm long, often elongating after flowering and as long as or slightly longer than peduncle, number of spikes per vertical shoot 1–4. No. of flower whorls 15–60 (20), flowers mostly with incomplete gynoeceum, no. of carpels variable, even within one plant, mostly 1, 1–2, or 2, rarely 1–3,

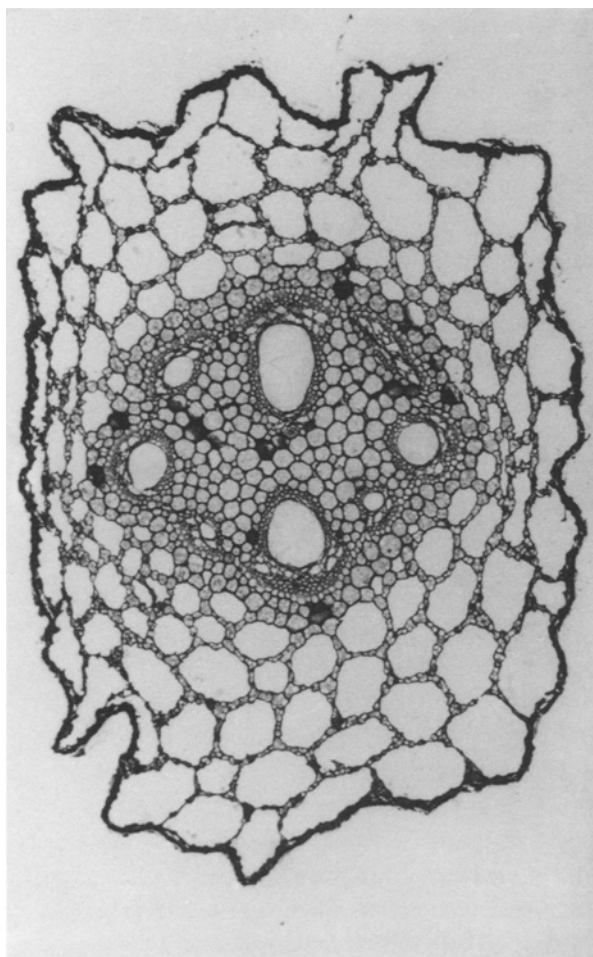


Fig. 1

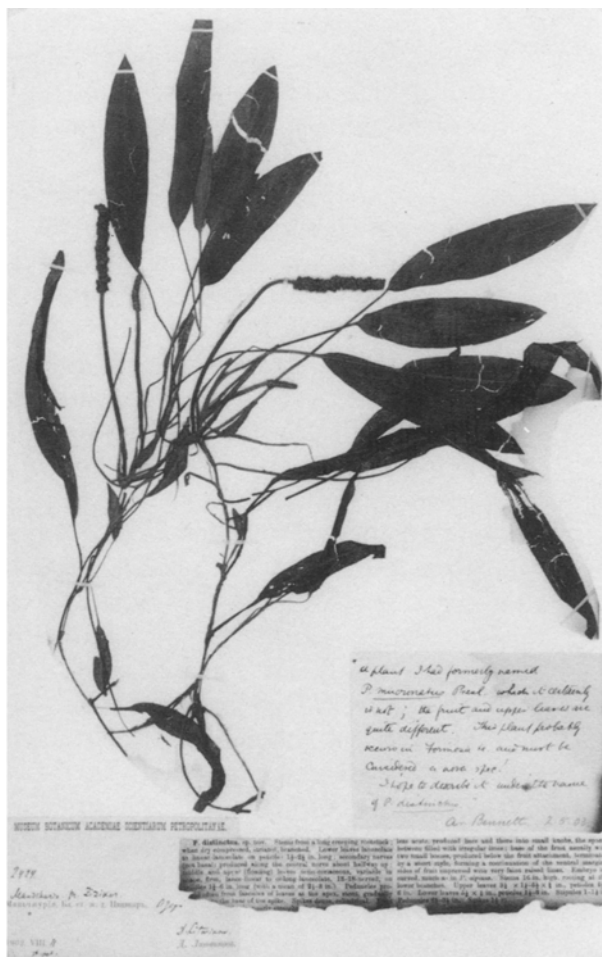


Fig. 2

Figs. 1 and 2. *Potamogeton distinctus*.—Fig. 1. Cross section of the stem, KADONO & WIEGLEB 896, OLD. — Fig. 2. Type specimen, LITWINOW 2424, BM

2–3, 3, or even 1–4. Length of ripe fruit 3–3.5 mm, variable in shape, roundish, with 1 to 3 more or less distinct keels.

Anatomy of the stem (Fig. 1): Stele of the eight bundles type, often with 2 phloema on the trio bundle (see also KADONO 1984: Fig. 11 c), median bundles relatively large, number of lateral bundles three, in rare cases four, sclerenchyma caps present, endodermis of O-type, interlacunar bundles always absent, subepidermal bundles often absent, rarely present, pseudohypodermis mostly absent.

Chromosome number 52 (HARADA 1956, TAKUSAGAWA 1961).

Variation. *P. distinctus* is characterized by a low degree of geographical variation, although, however, by a very high degree of polymorphism based on phenotypic plasticity. Polymorphism is not only expressed in the predominant growth form but also in size of the shoot complexes and vertical shoots, both in space and time. As there exist only few good icones of the species (p.e. Botanical Institute 1983), three different forms are shown in Figs. 2–4. Figure 2 shows the lectotype specimen

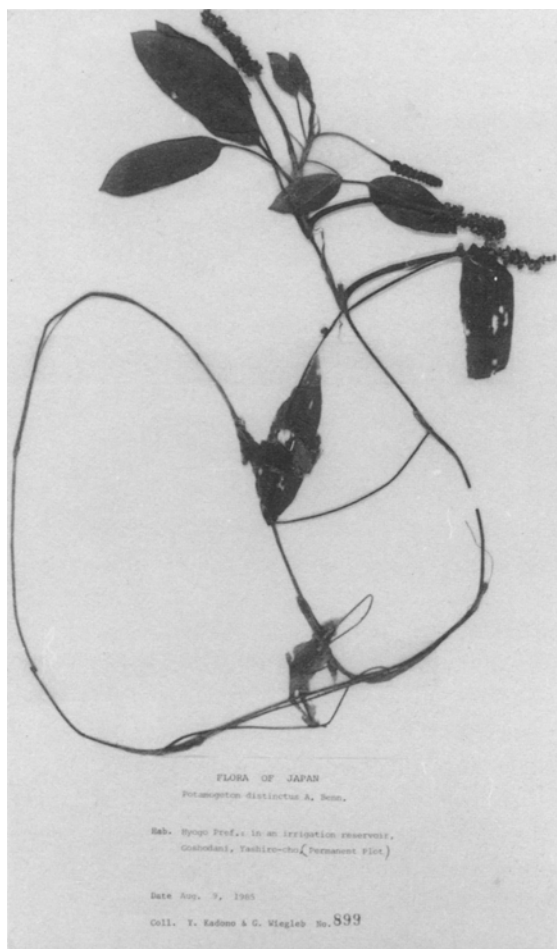


Fig. 3

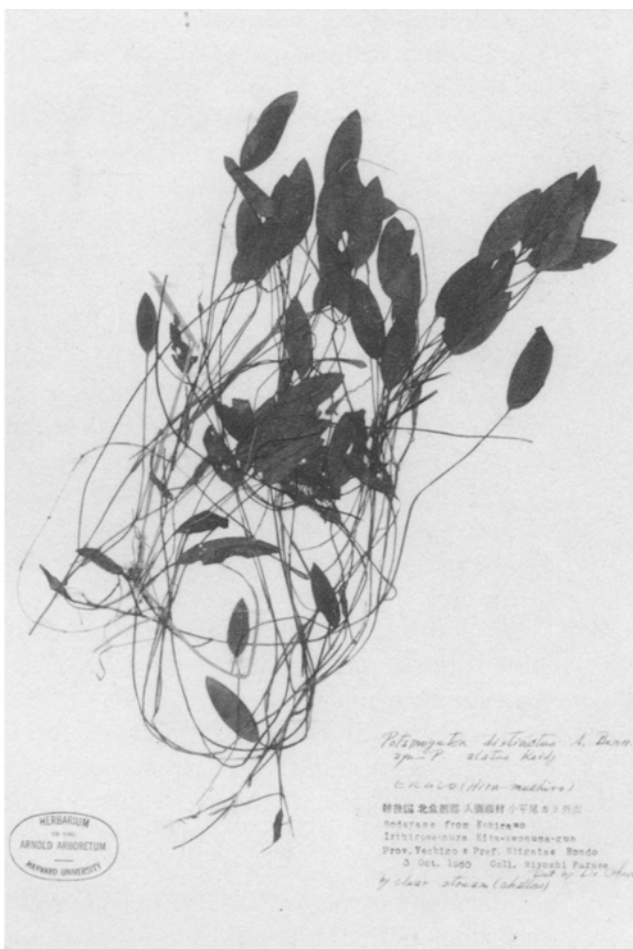


Fig. 4

Figs. 3 and 4. *Potamogeton distinctus*. — Fig. 3. Shallow water form, KADONO & WIEGLEB 899, OLD. — Fig. 4. Deep water form, FURUSE s.n., A

(LITWINOW 2424, BM), the exact habitat of which is unknown. Figure 3 shows a form from an irrigation reservoir, Fig. 4 one from a rivulet, both from Japan. The main growth forms are:

Land form: Short vertical shoots with coriaceous floating-leaf like leaves, vertical shoots born from a creeping stem that still has some contact to ground or surface water. It is found at the edges of the waters colonized or after desiccation of the whole habitat. The leaves stand erect in case of slow desiccation or are attached to the ground in case of quick desiccation.

Shallow water form: Vertical shoots up to 50 cm long, usually producing only a few submersed leaves, floating leaves abundant. It is found in ponds, pools and rice fields.

Deep water form: Vertical shoots more than 1 m long, heterophyllous, with both persisting submersed leaves and floating leaves. It grows in lakes and rivers.

Differential characters and affinities to other species. *P. distinctus* has been frequently confused with other species, belonging to no less than 6 different subsections or species groups, mainly with *P. polygonifolius* POURR., *P. natans* L. (incl. *P. morongii* BENNETT), *P. fryeri* BENNETT (incl. *P. insularis* NAKAI), *P. wrightii* MORONG (mainly as *P. malaianus* auct., also *P. tonkinensis* CAMUS), *P. sumatranus* MIQUEL, *P. gramineus* L. (incl. *P. gracilis* WOLFGANG), *P. tricarinatus* MUELLER & BENNETT (as *P. tepperi* BENNETT), and *P. nodosus* POIRET (incl. *P. indicus* ROXBURGH, *P. fluitans* auct., *P. occidentalis* CHAM. & SCHL., *P. delavayi* auct. and *P. tepperi* auct.).

The most distinctive character of *P. distinctus* is the presence of only 1 or 2 carpels in the typical form. There is, however, some indication of clinal variation in carpel number, because in Japan often only 1 or 1–2 carpels are observed, while on the continent 2 carpels are the rule and 1–3 or 2–3 are no rare exceptions (see for example the type specimen LITWINOW 2424, also Botanical Institute 1983). This leads to some difficulties with the delimitation against *P. nodosus* in the areas of geographical overlap. One must bear in mind that also in other species (p.e. *P. illinoensis* MORONG, see CORRELL & CORRELL 1982) plants with complete and incomplete (2 carpels) gynoeceium occur.

P. polygonifolius differs from *P. distinctus* by the regular presence of subepidermal bundles, the presence of 4 carpels, the smaller fruits, and the absence of specialized turions.

P. natans differs from *P. distinctus* by the presence of 4 carpels, the presence of interlacunar and subepidermal bundles, the presence of phyllodia in developed plants, the absence of turions, and the larger stipules.

P. fryeri differs from *P. distinctus* by the presence of 4 carpels, the long conical reddish fruit, the winged petioles of the floating leaves, the long whitish stipules, the different shape of the submersed leaves, and the presence of both interlacunar and subepidermal bundles.

P. tricarinatus s.l. differs from *P. distinctus* by the more leathery floating leaves, which are often cordate at the base, the more prominent tricarinate fruits, and the completely different stem anatomy (strong sclerenchyma in the stele, large number of interlacunar bundles).

P. wrightii differs from *P. distinctus* by the presence of 4 carpels (but see below, *P. × malainoides*), the acutely tipped submersed leaves, the more tuber-like winterbuds, and the presence of interlacunar bundles. *P. sumatranus* and *P. gramineus* can only be confused with *P. distinctus* in the state of landforms.

P. nodosus is the species most similar to *P. distinctus* which is mainly expressed by the common turion shape and the similarity of stem anatomical characters. The main difference is that *P. nodosus* usually has 4 carpels when flowering some of which may later remain undeveloped. *P. nodosus* can be found with almost unpetioled submersed leaves while the petioles of *P. distinctus* are longer (also in floating leaves). *P. nodosus* in most cases is opportunistic in flower insertion (with spikes being inserted already at very low nodes of the vertical shoot), while *P. distinctus* is mainly, but not always, deterministic (with flowers being regularly inserted at nodes of higher number, WIEGLEB & KADONO 1988). All differences, however, are more or less gradual and *P. distinctus* may be included in *P. nodosus* as a subspecies in future studies (see also below, discussion on *P. indicus* and *P. delavayi*).

ASCHERSON & GRAEBNER (1907) treated *P. distinctus* together with *P. wrightii* (as *P. malaianus*) and *P. sumatranus* within the *Lucentes* (= *P. lucens* group, WIEGLEB 1988). As GRAEBNER relied very much on the determinations and descriptions of BENNETT, many specimens nowadays regarded as *P. distinctus* were treated in other subsections. Unfortunately, HAGSTROEM (1916) was not familiar with the species. With the material available to him he was unable to recognize the relations to *P. franchetii* etc. He only expressed some doubt about the inclusion into the *Lucentes*. Considering the facts stated above *P. distinctus* belongs to the *P. nodosus* group (WIEGLEB 1988; subsect. *Nodosi*, HAGSTROEM 1916).

Distribution. The analysis of the distribution is based only on specimens that can be identified and localized with a certain degree of exactness.

U.S.S.R.: Not seen. There are, however, several quotations of specimens that might refer to the species under study: East Siberia: YUZEPCHEK (1968) mentions *P. franchetii*. BENNETT (1914) refers to this specimen as *P. perversus*. — Far East: YUZEPCHEK (1968) mentions *P. franchetii*, *P. distinctus* and *P. digynus*. — Kamtschatka: *P. franchetii* is mentioned by HULTEN (1927), MIYABE & KUDO (1931) and YUZEPCHEK (1968). Questionable. In S, only HULTEN specimens belonging to *P. natans* are preserved. — Saghalien: See MIYABE & KUDO (1931), as *P. franchetii*. — Kurile Islands: See MIYABE & KUDO (1931). Questionable. OHWI 669 (UPS) is *P. natans*.

Korea: Mainland: FAURIE 34, 84 (P), 222 (BM, E), 223 (BM, E), 689 (E, ZT), 2080 (E), 10025 (BM). — Archipelago: OLDHAM 824 (A, GOET, ZT), s.n. (M).

China: Manchuria: LITWINOW 2322 (BM), 2424 (type locality, A, BM), 2445 (A, BM), 3351 (A, BM). — Southern and Central parts: ALLISON 214 (A), BODINIER 2442 (E), CHIAO 2543 (A, BM, E, F, MICH), CHOW 8382 (A), CHUNG 8 (A), CLEMENS 6054 (E), DAVID s.n. (P), ESQUIROL 456, s.n. (E), FANG 14584 (A), FARGES 650 (P), FORREST 1886 (BM), 2644 (E), 7040 (E), 8383 (E), s.n. (E), GLASSFIELD s.n. (CGE), HENRY 314 (MO), 2366 (A, E), 3603 (A, BM), 9985 (MO), LICENT 1295, 6425 (BM), MERRILL 11376 (A), SCHNEIDER 2219 (A), TAI T 935 (A), TING & SHI 1502 (AAU), TSIANG 5053 (A), WANG 23107 (A). — Tibet: HÜGEL 1281 (LD), LUDLOW & SHERRIFF 8818 (BM), LUDLOW, SHERRIFF & ELLIOTT 14216 (BM).

Japan: Distribution map in KADONO (1982). Only representative specimens are listed. Hokkaido: SAKUUCHI s.n. (PH), KADONO 748 (BRI). — Honshu: ARIMOTO s.n. (MO), FAURIE 29, 30, 31, 34b, 35 (WU), 36 (BM, WU), 39, 41 (WU), 107 (MO), FUKUOKA & KUROSAKI 4032 (TI), FURUSE s.n. (A, UC), KADONO 594, 595 (BRI), KADONO & WIEGLEB 3, 4, 895–900, 1085–1087, 1089–1091, 1093–1099 (OLD), KIMURA & KYOGOKU s.n. (A), KITAMURA s.n. (TI, ZT), MAKINO 226712, 226696 (E), MAXIMOWICZ s.n. (BM, WU), OHBA & al. 380118 (CHR), s.n. (A), OHWI & ITO 182 (A, B, BM, E, F, L, MICH, MO, UC, UPS, ZT), SAKURAI s.n. (E), SCHRÖTER s.n. (ZT), SHIMIZU 14383 (TI), TAKAHASHI & SASTROUTOMO 435 (MO), 436 (A, TI), WATANABE s.n. (A). — Sado Island: FAURIE 2450 (BM). — Shikoku: KRUG 1197 (B). — Kyushu: KADONO & WIEGLEB 1099 (OLD). — Ryukyu Islands: AMANO 7810 (TI), HATUSIMA 17610 (L, TI), 18034 (L), 18085, 18097 (TI), WALKER & al. 6079 (L, TI, UC).

Taiwan: FAURIE 532 (BM), OLDHAM 636 (A), SHIMADA 014282 (TAI).

Philippines: Luzon: Not seen. One specimen mentioned in BENNETT (1914) under *P. perversus* may refer to the species.

Vietnam: Not seen. All specimens studied which are mentioned by CUONG & VIDAL (1983) are doubtful. BON 3543 (= *P. tonkinensis* CAMUS, in P) is referred to *P. wrightii*. PETELOT 3747 (A, P) has four carpels (see figure in CUONG & VIDAL 1983). The specimen shows characters of both *P. nodosus* and *P. wrightii* (WIEGLEB 1990).

Thailand: KERR 6260 (AAU, BM).

Nepal: WALLICH s.n. (WU).

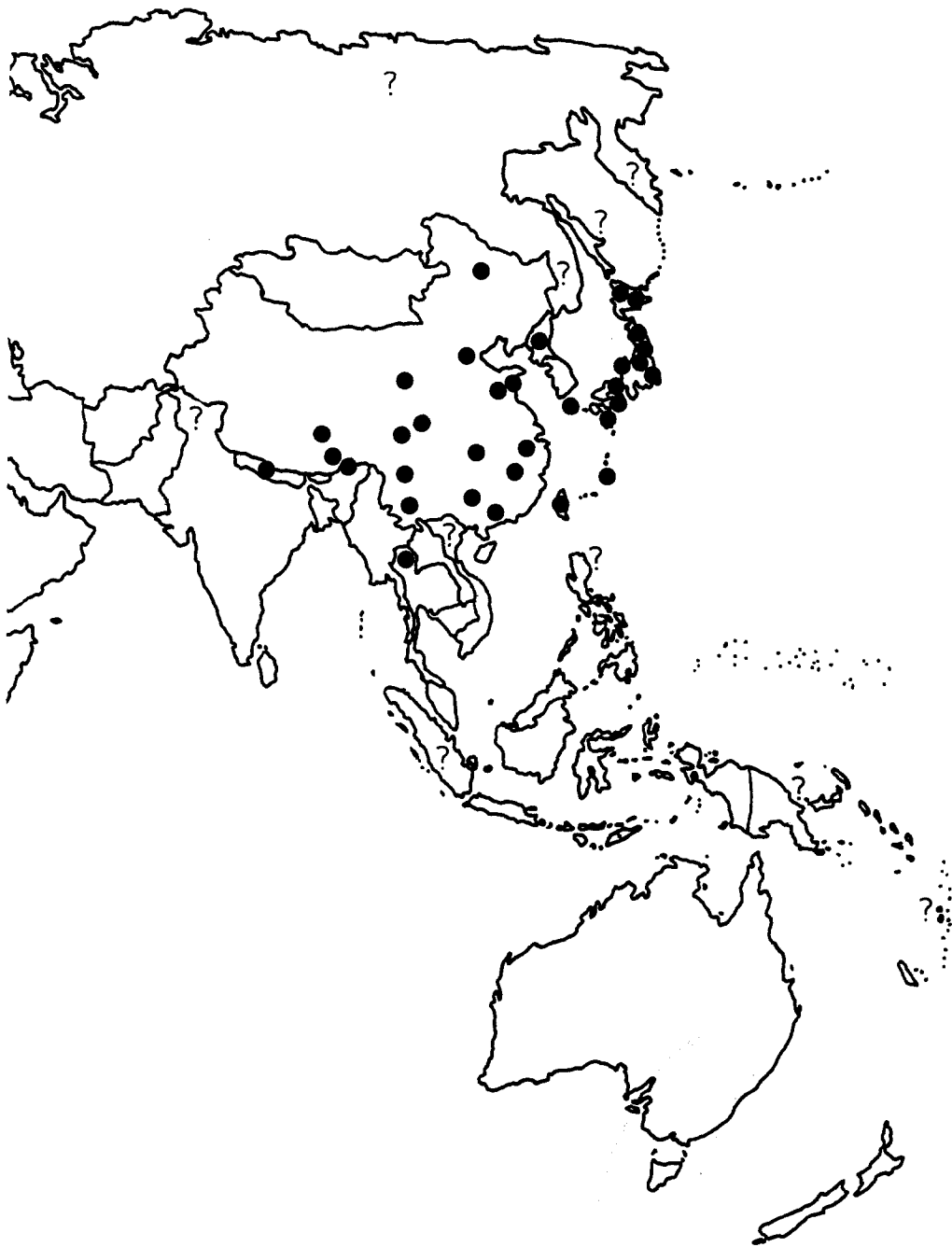


Fig. 5. Distribution map of *Potamogeton distinctus*

Bhutan: Difficult to decide. LUDLOW, SHERRIFF & HICKS 16997, 19586, 19636, 21020 (BM) come closest to *P. distinctus*.

A distribution map is shown in Fig. 5. The species seems to have a strict E. Asian distribution, but the exact limits are still unknown. From the northern part of the area no specimens were available. On the Asian mainland particularly in the Himalaya area there are problems with the delimitation to *P. nodosus*. In SE.

Asia records of broad-leaved *Potamogeton* are generally sparse and often not to identify with certainty.

The following specimens from several places outside the area of *P. distinctus* show similarities in some characters but an exact assignment is impossible:

India (Kashmir): LUDLOW & SHERRIFF 8085 (BM).

New Guinea: JERMY 3940 (L), RIFAI 6632 (L).

New Hebrides: MORRISON s.n. (A). The very long petioles of this specimen are reminiscent of *P. distinctus*.

Sumatra: The landforms ALSTON 14358 (BM), TEIJSMAN s.n. (BREM, L, U), ZOLLINGER 3787 (BM) are similar to landforms of *P. distinctus*, but may likewise belong to *P. nodosus* or *P. sumatranus*.

Ecology. *P. distinctus* is found in different kinds of habitats. According to observations in Japan (WIEGLEB & KADONO 1988, 1989) and also herbarium labels it is most frequent in artificial ponds and reservoirs. It is also frequent in other kinds of shallow waters like rice fields, irrigation ditches and even swamps. On the other hand it is also found in rivers and lakes in a depth of more than one meter.

The growth of the species is seasonal and starts in April with germination either from overwintering turions or seeds. It is able to endure longer lasting droughts by means of a seed bank. *P. distinctus* is almost always producing numerous fertile fruits. Reduction of carpels obviously leads to a higher probability in seed production which may be seen as an adaption to regular sexual reproduction in a very unpredictable environment (e.g., rice fields and irrigation ponds which are subjected to frequent desiccation).

Besides that the formation of tubers regularly occurs which can be induced both by drought and low temperatures. Vegetative reproduction by means of fragmentation is not effective in this species (KADONO 1984). Shorter periods of drought are tolerated by the formation of landforms (see above). The species can endure a high degree of natural and anthropogenic disturbance and withstands even intensive weed control in rice fields (KADONO 1982).

The current velocities in rivers and rivulets may range up to 0.6 m/sec. Both sandy and muddy substrates are colonized. KADONO (1982) reports the following ranges of some important waterchemical parameters (minimum – maximum/median): pH 6.1 – 10.9/7.3; alkalinity (mmol/l) 0.08 – 1.66/0.48; Calcium (mg/l) 1.12 – 24.36/7.05; chloride (mg/l) 4.1 – 202/11.8; conductivity (µS) 15 – 815/86. Relative to most other Japanese species, *P. distinctus* has a certain relation to low alkalinity and soft water conditions. But it does not occur in truly acidic or dystrophic water (like *P. fryeri*). The high pH indicates some degree of eutrophication in poorly buffered waters. The species seems to tolerate some pollution in the riverine habitat, where it is frequently associated with *P. octandrus* (KADONO 1982, WIEGLEB & KADONO 1988). It is most frequent in lowland areas, but is also found in mountains up to an altitude of 3 600 m (in the Himalaya area).

Hybrids. Even though *P. distinctus* is frequently sexually reproducing only one doubtful hybrid has been reported.

P. × malainoides MIKI, Water Phan. Japan: 20. 1937 (= *P. distinctus* × *P. wrightii*); OHWI, Fl. Japan: 121. 1984. Type: Lake Biwa. MIKI (OSA).

The identity of *P. × malainoides* is still uncertain. The plants from the type locality may also represent floating leaved forms of *P. wrightii* (Y. KADONO, pers. comm.). I have seen landforms with coriaceous leaves from the type locality. In this case the name must be considered as a synonym of that species (see WIEGLEB 1990).

On the other hand there exist plants with intermediate characters between the parent species having 2 to 3 developed carpels (by which they are distinguished from *P. wrightii* with 4 carpels) and having scattered interlacunar bundles (by which they are distinguished from *P. distinctus* having no such bundles). Examples have been listed in WIEGLEB (1990).

2. The identity of *P. tepperi*, *P. indicus*, and *P. delavayi*

***Potamogeton tepperi* BENNETT.** *P. tepperi* was first described from Australia by BENNETT (1887) who indicated affinities to *P. natans*, *P. polygonifolius* and *P. leschenaultii*, the latter one being a dubious synonym of *P. nodosus*. Later BENNETT (1896) applied the name *P. tepperi* to plants from outside Australia covering the large geographical area between India, Indonesia and Japan. However, BENNETT (1914) himself and later MIKI (1937) doubted the identity of *P. tepperi* over such a large geographical range. HAGSTROEM (1916) treated the species among the *Amplifolii*, but the stem anatomical characters mentioned by him indicate some affinity to the *Muricati* (= *P. tricarinatus* group).

In Australia, *P. tepperi* is still recognized in some parts (EICHLER in Flora of S. Australia, fide ASTON 1973), but synonymized with *P. tricarinatus* in others (ASTON 1973). The name is used by KAK (1983) in the Himalaya region for a *P. nodosus*-like plant. MENDOZA & DEL ROSARIO (1967) include both *P. tepperi* and *P. perversus* in their treatment of Philippine *Potamogeton*. This is based on the introduction of *P. perversus* by BENNETT (1914), which is a mixture of specimens belonging to different taxa (*P. distinctus*, *P. nodosus* and *P. natans*-hybrids).

The following specimens have been identified at least by one researcher as "*P. tepperi*" on herbarium labels. Often besides *P. tepperi* also other guesses were found. The specimens are assigned here as follows:

P. tricarinatus complex: Australia: WALTER 2464 (B), JACKSON 64, ADWAY 4648, GRAY 5790, EICHLER 13838, BARKER 713, MEYERS s.n. (CANB) BAILEY 958 (C). *P. tepperi* must be regarded as a valid taxon within this complex. It is not a simple synonym of *P. tricarinatus*. Further research is needed.

P. drummondii: Australia: Kangaroo Island, EICHLER 15487 (CANB).

P. nodosus: China: DUCLOUX 2571. — Timor: WALSH 283 (BM), cf. also VAN STEENIS 18113 (L, as *P. malayanus*). — Philippines: MERRILL 1722 (A, BM, L, QU), ROBINSON 14110 (BM, F). *P. tepperi* from the Philippines is a distinct form of *P. nodosus*. It has roundish floating leaves and was often collected without submersed leaves. The identity of the specimens with *P. nodosus* is verified by the specimen ELMER 5952 (in P, identified as *P. gaudichaudii*, see Fig. 6) which is from the same locality. So far no authentic specimen of *P. perversus* from the Philippines could be studied to see whether there is another broad-leaved species present.

P. distinctus: All specimens from Korea, China (except Yünnan), Japan and Taiwan. Furthermore from Thailand: KERR 6260 (AAU, BM).

***Potamogeton indicus* ROXBURGH.** *P. indicus* was described by ROXBURGH (1820) from India. ASCHERSON & GRAEBNER (1907) adopted the concept of *P. indicus* while assigning specimens to this species that from a modern point of view belong to either *P. nodosus* POIRET or *P. distinctus* BENNETT. HAGSTROEM (1916) considered *P. indicus* as a doubtful synonym of *P. nodosus*. But he also pointed out that besides plants that are indistinguishable from *P. nodosus* specimens have been found that considerably differ in their stem anatomy.

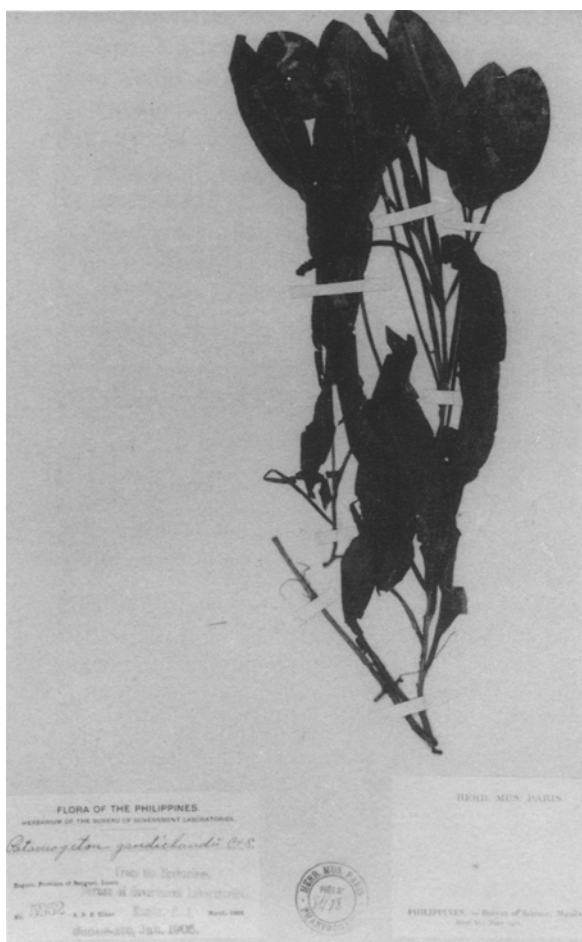


Fig. 6

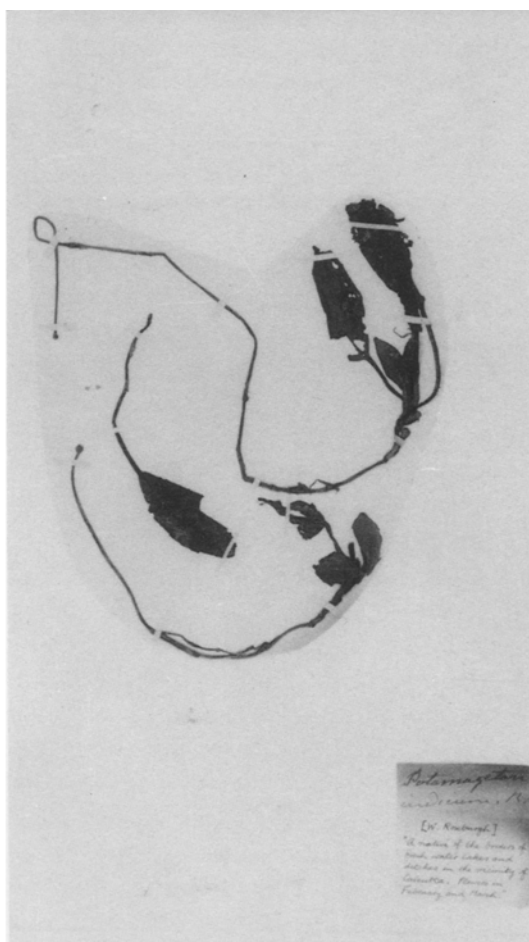


Fig. 7

Fig. 6. *Potamogeton* “*tepperi*” from the Philippines, ELMER 5926, P, as *P. “gaudichaudii”*.—Fig. 7. The supposed type specimen of *P. indicus* ROXBURGH, BM, collected in Calcutta Botanical Garden

Modern treatment of *P. indicus* is inconsistent. BOKHARI & AFZAL (1979) do not mention it at all. BISWAS & CALDER (1955) name it as the only broad-leaved pondweed from the Indian subcontinent. SHARMA & CHATTERJEE (1967) and MISRA (1972) follow this nomenclature. As they describe strong variation in chromosome number, their findings are quite inconsistent with the idea of a single taxon *P. indicus*. DANDY (1971, 1978) synonymizes it with *P. nodosus*. KAK (1983) follows the synonymization but creates further confusion by introducing the taxa *P. polygonifolius* (which does not occur in India), *P. tepperi* (see above) and *P. fluitans* (which is an obsolete synonym of *P. nodosus*).

The specimens mentioned in the following were assigned to *P. indicus* by at least one researcher. They would be identified as follows:

P. wrightii: Bangla Desh: SHAIK MOKIM s.n. (AAU). — Kashmir: SCHLAGINTWEIT 4615 (BM, type of *P. hindostanicus* HAGSTROEM).

P. nodosus: India: ROXBURGH s.n. (the "type" or at least from the type locality, BM, see Fig. 7), this plant has repeatedly been recollected (anonymous s.n., L; Herb. Drake s.n., F).

All other specimens studied; furthermore all specimens named *P. "roxburghianus"* SCHULTES. Kashmir: STEWART s.n. (B), KOELTZ 9039 (B). — Nepal: RAM 199 (A). — Bangla Desh: SHAIK MOKIM s.n. (A). — Sri Lanka (also as *P. natans*): THWAITES 590 (CGE, GOET, A, BM, BREM). — Burma: AUBERT & GAGE s.n. (ABD). — Andaman Islands: PRAIN s.n. (L, Z).

P. distinctus: Japan: FAURIE 34 (P).

As a consequence *P. indicus* has to be considered as a synonym of *P. nodosus*, which was also misapplied to *P. distinctus* and *P. wrightii*. No convincing differential characters could be found so far. Indian *P. nodosus* sometimes develops completely unpetioled submersed leaves and peduncles opposite to alternate leaves. The last named character was already included by ROXBURGH in his description. Both characters were not yet observed in *P. nodosus* from outside India. Thus it may be justified to retain the taxon *P. indicus* on varietal rank, as such forms cooccur with typical *P. nodosus*.

Many specimens of Indian *P. nodosus* have been collected as *P. "natans"*. On the other hand the specimen POLUNIN & al. 5156 (A, E, as *P. nodosus*) seems to be more likely *P. natans*. *P. natans* does occur in Kashmir, Nepal, Bhutan and Tibet, but also south of the Himalaya Mountains, as most probably in the Khasia Hills. The exact southern limit is unknown, but in the northern part of India confusion between *P. nodosus* and *P. natans* may have caused the irritations mentioned by HAGSTROEM (see above).

There remain some Indian plants with unknown affinity, as for example: SANTAPAN 16613 (MO) and HAMILTON 449/1 (E), with submersed leaves only, but flowering. This was not observed in other specimens of *P. nodosus*.

There remain also some plants from Bhutan and adjacent parts of Tibet with uncertain affinity (all in BM): LUDLOW & SHERRIFF 8672, 9451, 9563, 10072; LUDLOW, SHERRIFF & HICKS 20149; LUDLOW, SHERRIFF & ELLIOTT 12570, 14198, 14228. — None of these plants belongs to the *P. nodosus* group. 9451 is more *P. natans*-like, 10072 and 20149 are more *P. gramineus*-like.

***Potamogeton delavayi* BENNETT.** The name *P. delavayi* BENNETT originally refers to the specimen DELAVAY 4750 (in P). This is a slender plant similar to *P. distinctus*. As flowering is already over, one cannot decide whether half-ripe fruits may have fallen from the inflorescence, or whether the gynoeceium has been incomplete during anthesis. Thus the name may refer either to *P. distinctus* or *P. nodosus*.

Later on it has also been applied in herbaria to *P. nodosus*-like plants from the same area. FENG 1354 (A), YÜ 12450 (A), DELAVAY 3844 (P), and DUCLOUX 2571 (P, which was included into the description of *P. tepperi* var. β *subcordatus*, CAMUS 1910) can better be regarded as *P. nodosus*. Also the figure in Botanical Institute (1983) with 3 fully developed and 1 partly developed carpel points to *P. nodosus*.

It cannot be ruled out that *P. delavayi* is an older synonym of *P. distinctus* and may thus have priority. But it cannot be proven, either, with the specimen in P. Whatever the status of this form is, one cannot maintain *P. delavayi* as a separate species between *P. nodosus* and *P. distinctus*.

Conclusions

It can be seen that within the *P. nodosus* group (subsect. *Nodosi* ss. HAGSTROEM) there is no room for more than one species (*P. distinctus*) besides the very variable *P. nodosus*. The situation is still dissatisfying as the variation among the various local and regional forms of *P. nodosus* is enormous. *P. distinctus* can only be upheld as a distinct taxon if it can be shown that despite all morphological fuzziness it has a recognizable area, and does not represent a collection of long-petioled, seed producing ecotypes of *P. nodosus* which live under special ecological conditions.

So far no systematic treatment of *P. nodosus* has been carried out. However, many specimens of this species have been seen during the revision of *P. distinctus* (the present paper) and *P. wrightii* (WIEGLEB 1990). The area of *P. nodosus* is greater than formerly assumed (HAGSTROEM 1916, OGDEN 1943, YUZEPCHEV 1968, SYMOENS & al. 1979, CASPER & KRAUSCH 1980). In northern Asia it occurs in W. Siberia. In C. Asia it is present in Pakistan, Afghanistan, and the Pamir-Alai mountains. It may be present in northern China (LICENT 1295 in BM). It is present in Kashmir and Nepal, where also *P. distinctus* does occur. Quite puzzling is the situation in Yunnan where also both species seem to occur together.

In southern Asia *P. nodosus* occurs on the whole Indian subcontinent, on Sri Lanka, the Andaman Islands, Burma, Thailand (CHAI-ANAN 1985 as *P. "polygonifolius"* POURR.) and most probably Vietnam. In the Australasian Islands it is present on Sumatra (SCHIFFNER 1477, A), Java (ZOLLINGER 3778, BM), Celebes (TEJSMAN s.n., BREM, L, U), Luzon (ELMER 5952, P), and probably Alor (JAAG 1014, ZT). In the Pacific area *P. nodosus* not only occurs on Hawaii, but also on the Marianes (THOMPSON 231, BM), New Caledonia (HARTLEY 14978, CANB; McKEE 8151, L), and the New Hebrides (RAYNAL 16393, A). The occurrence on New Guinea and on the Australian mainland (BEAGLEHOLE ACB 27506, MO) is questionable.

A difficult situation is found on Taiwan. There, in one of the centres of the occurrence of *P. distinctus* also 4-carpelled *P. nodosus*-like plants are found (DE VOL 9014, PH). YANG (1978) referred to this specimen as *P. distinctus*, even though more recently he also recognized *P. nodosus* as present in E. Asia (in herb. MO). Especially this finding indicates that the taxonomy of E. Asian broad-leaved pondweeds is far from being understood.

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